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ASSIGNMENT 12

Problem Statement: Write a program in C to accept two numbers from the user & compute:
a) The smallest common divisor of two numbers.
b) The Greatest Common Divisor (GCD) of two numbers

Objective:
- To understand the use of mathematical operations in C
- To implement loops and conditional statements
- To perform number-based computations.
- To enhance logical thinking using menu-driven programs.

Theory: Divisors & the Greatest Common Divisor (GCD) are fundamental concepts in number theory. The smallest common divisor of two numbers is the smallest positive integer greater than 1 that divides both numbers.

Platform: Linux

Algorithm:

Step 1: START.
Step 2: Read two integers a & b from the user.
Step 3: Find the smallest common divisor.
- set a flag found = 0
- for i from 2 to min(a,b);
- If a % i == 0 & b % i == 0;
  print i as the smallest common divisor.
- set found = 1
- Break the loop.

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- If found==0, print "No common divisor other than 1"

Step 4: Find the GCD
- while b≠0:
  temp=b b=a%b a=temp.
- The value of a is the GCD

Step 5: Display the GCD
Step 6: STOP.

C Program:

```
# include <stdio.h>
int main() {
    int a,b, i, min;
    int found=0;
    printf(" Enter 2 num:");
    scanf(" %d %d", &a, &b);
    min=(a<b)? a:b;
    for(i=2; i<=min; i++){
        if(a%i==0 && b%i==0){
            printf(" Smallest divisor: %d\n",i);
            found=1;
            break;
        }
    }
    if (!found){
        printf(" No common divisor", i);
        found=1;
    }    break
    int x=a, y=b, temp;
    while(y!=0) {
        temp=y;
        y=x%y;
```

```
        x = temp;
    }
    printf("GCD of two numbers: %d\n", x);
    return 0;
}
```

Input:
The user enters two integer numbers.
Example input: Number 1: 24 Number 2: 36

Output: The program displays the smallest common divisor and the GCD of two numbers.

Ex: Output: Smallest Common Divisor: 2
Greatest Common Divisor (GCD): 12

Conclusion: Thus, the program to find the smallest common divisor & the Greatest common Divisor of two numbers was successfully implement using C programming.

FAQS:

1. What is meant by the Greatest Common Divisor (GCD)?
A. The GCD of two or more integers is the largest positive number that divides each of them exactly, leaving no remainder.

2. Why does the Euclidean algorithm efficiently compute the GCD?
A. The Euclidean algorithm efficiently computes the GCD because it repeatedly reduces the problem size.

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- It uses the rule:
  GCD(a,b) = GCD(b, a%b)

3. How can this program be modified to handle negative numbers?

A. a) Convert inputs to positive values using abs()
b) Then perform divisor & GCD calculations.
c) GCD is always taken as a positive number.
d) Negative signs do not affect divisibility.

4. What changes are required to compute the Least Common Multiple (LCM) using the GCD?

A. a) First compute the GCD
b) Use the formula. LCM = (a*b)/GCD
c) Use absolute values to handle negative inputs safely.

5) How can this program be extended to find the LCM of two numbers?

A. a) Compute the GCD using the Euclidean algorithm.
b) Apply the LCM formula.

$$LCM = \frac{|a \times b|}{GCD}$$

Code:

```
int lcm;
lcm = (a*b)/gcd;
printf("LCM of 2 num: %d\n", lcm);
```